because, unlike what we found using the laws of classical physics, splattering solves the full quantum equations but unsplattering doesn't. A reverse-run movie of a splattering egg would then depict motion that couldn't happen in the real world, which would explain why we've never seen it. And that would be that.

Possibly. But even though this would seem to provide a very different explanation of time's arrow, in reality it may not be as different as it appears. As we emphasized in Chapter 6, for the pages of *War and Peace* to become increasingly disordered they must begin ordered; for an egg to become disordered through splattering, it must begin as an ordered, pristine egg; for entropy to increase toward the future, entropy must be low in the past so things have the potential to become disordered. However, just because a law treats past and future differently does not ensure that the law dictates a past with lower entropy. The law might still imply higher entropy toward the past (perhaps entropy would increase asymmetrically toward past and future), and it's even possible that a time-asymmetric law would be unable to say anything about the past at all. The latter is true of the Ghirardi-Rimini-Weber proposal, one of the only substantive time-asymmetric proposals on the market. Once their collapse mechanism does its trick, there is no way to undo it—there is no way to start from the collapsed wavefunction and evolve it back to its previous spread-out form. The detailed form of the wavefunction is lost in the collapse—it turns into a spike—and so it's impossible to "retrodict" what things were like at any time before the collapse occurred.

Thus, even though a time-asymmetric law would provide a partial explanation for why things unfold in one temporal order but never in the reverse order, it could very well call for the same key supplement required by time-symmetric laws: an explanation for why entropy was low in the distant past. Certainly, this is true of the time-asymmetric modifications to quantum mechanics that have so far been proposed. And so, unless some future discovery reveals two features, both of which I consider unlikely— a time-asymmetric solution to the quantum measurement problem that, additionally, ensures that entropy decreases toward the past—our effort to explain the arrow of time leads us, once again, back to the origin of the universe, the subject of the next part of the book.